C. Creating Keys for StORages Has Become My Main Skill

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1200 | Tags: bitmasks, constructive algorithms, greedy | https://codeforces.com/problemset/problem/2072/C

Akito still has nowhere to live, and the price for a small room is everywhere. For this reason, Akito decided to get a job at a bank as a key creator for storages.

In this magical world, everything is different. For example, the key for a storage with the code (n, x) is an array a of length n such that:

- $a_1 \mid a_2 \mid a_3 \mid \ldots \mid a_n = x$, where $a \mid b$ is the bitwise "OR" of numbers a and b.
- MEX({ a_1, a_2, a_3, …, a_n })□ is maximized among all such arrays.

Akito diligently performed his job for several hours, but suddenly he got a headache. Substitute for him for an hour; for the given n and x, create any key for the storage with the code (n, x).

□MEX(S) is the minimum non-negative integer z such that z is not contained in the set S and all $0 \le y$ lt z are contained in S.

Input

The first line contains the number t ($1 \le t \le 10^4$) — the number of test cases.

In the only line of each test case, two numbers n and x ($1 \le n \le 2 \cdot 10^5$, $0 \le x$ lt 2^{30}) are given — the length of the array and the desired value of the bitwise "OR".

It is guaranteed that the sum of n across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output n integers a_i ($0 \le a_i$ lt 2^{30}) — the elements of the key array that satisfy all the conditions.

If there are multiple suitable arrays, output any of them.

Example

Input
9 1 69 7 7 5 7 7 3 8 7 3 52 9 11 6 15 2 3

Output

```
69
6 0 3 4 1 2 5
4 1 3 0 2
0 1 2 3 2 1 0
7 0 6 1 5 2 4 3
0 52 0
0 1 8 3 0 9 11 2 10
4 0 3 8 1 2
0 3
```